1 Two identical coils are connected together and arranged as shown in Fig. 1.1.

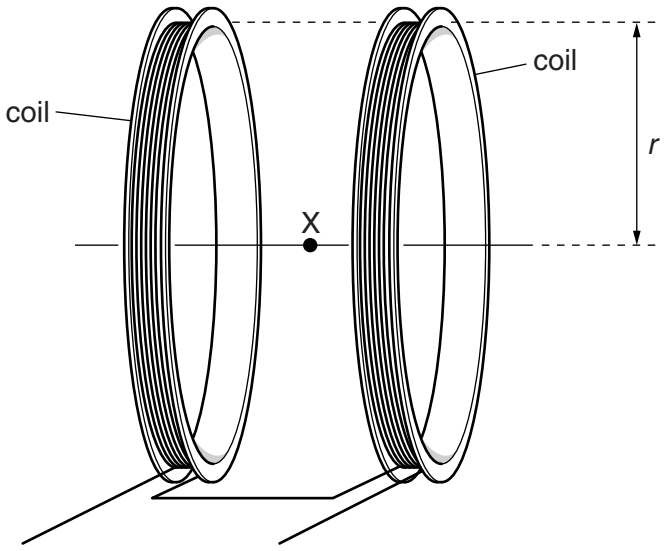

Fig. 1.1

The coils are in the vertical plane and are parallel to each other. When the coils are connected to a power supply, there is a magnetic field between them.

It is suggested that the magnetic flux density B of the field at the point X is related to the radius r of the coils by the relationship

$$B = \frac{0.72\mu_0 NI}{r}$$

where N is the number of turns on each coil, I is the current in the coils and μ_0 is the permeability of free space.

Design a laboratory experiment that uses a Hall probe to test the relationship between B and r and determine a value for μ_0. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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